Mehul Parmar

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PROFILE SUMMARY:

Dynamic and versatile professional with over two years of experience in software development and data science. Expertise in Python, Django, and Machine Learning, with a strong foundation in Artificial Intelligence and a proven track record in building scalable web applications and deploying machine learning models. Proficient in tools and frameworks like Scikit-learn, Pandas, SQL, and NodeJS, with a passion for creating innovative solutions that drive business impact. Known for collaborative teamwork, problem-solving, and delivering secure, efficient software systems to meet diverse organizational needs.

EDUCATIONAL QUALIFICATION:

Long Island University, Brooklyn, NY — May-2025
MS Artificial Intelligence
GPA: 3.9 / 4.0

Gujarat Technological University, Gujarat, India — June-2022
Bachelor's of Engineering, Information Technology
GPA: 3.55 / 4.0

TECHNICAL SKILLS:

- **Programming Languages:** Python, Data Science, HTML, CSS, JavaScript
- **Frameworks:** Bootstrap, Django, NodeJS
- **Machine Learning & AI**: Scikit-learn, PyTorch, OpenCV, Pandas, NumPy
- **Databases:** MYSQL, SQL, SQLite
- **Version Control:** GitHub, Git
- **OS:** Windows, Linux
- **Soft Skills:** Communication skills, Problem-solving, Time management, Leadership, Team player, Interpersonal skills.

PROFESSIONAL EXPERIENCE:

Python Developer — August 2021-May 2023
BrainyBeam Technologies Pvt. Ltd. (Ahmedabad, GJ, India)

- Writing efficient, reusable, testable, and scalable code
- Understanding, analyzing, and implementing – Business needs, feature modification requests, and erosion software components
- Developing – Backend components to enhance performance and receptiveness, server-side logic, and platform, statistical learning models, and highly responsive web applications
- Designing and implementing – High availability and low latency applications, data protection, and security features
- Performance tuning and automation of the application
- Working with Python libraries like Pandas, NumPy, etc.
- Creating predictive models for AI and ML-based features
- Implement security and data protection solutions
- Coordinate with internal teams to understand user requirements and provide technical solutions

PROJECTS:

Medical Image Analysis | Code Spring2024-LIU

- I spearheaded a project focused on medical image analysis, leveraging deep learning techniques and aiming to enhance clinical diagnosis, treatment planning, and disease management.
- Developing robust models utilizing Convolutional Neural Networks (CNN), Random Forest and Support Vector Machines (SVM).
- I achieved impressive accuracies of 86%, 85% and 81%, respectively.
- By employing Python and Jupyter Notebook, along with essential libraries such as NumPy, PyTorch and OpenCV, I meticulously crafted classification reports, confusion matrices, ROC curves and AUC scores for comprehensive model evaluation.
- With a primary focus on brain images derived from CT scans and X-rays, the project aimed to significantly improve diagnostic precision and reduce interpretation time, thereby elevating the standard of patient care within clinical environments.

Autism Spectrum Disorder Detection | Code Spring2024-LIU

- Conducted research focusing on leveraging machine learning techniques for Autism Spectrum Disorder(ASD) detection, aiming to enhance screening efficiency and diagnostic accuracy.
- Utilizing Python and Jupyter Notebook, I explored various classification algorithms including Decision Trees, Random Forest, Support Vector Machines (SVM), K-Nearest Neighbors (K-Neighbors), Logistic Regression and Mutli-layer Perceptron.
- I meticulously evaluated model performance using classification reports, confusion matrices, ROC curves, AUC scores, and F-beta scores.
- By addressing the shortcomings of existing ASD screening tools, particularly in alignment with DSM-5 criteria, the project aimed to contribute toward more reliable and effective diagnostic methodologies.

Android Malware Detection Model | Code Fall2023- LIU

- I have developed an Android Malware detection model using Python and Machine Learning principles.
- The system analyzed mobile application characteristics to differentiate between benign and malicious software.
- Using various machine learning algorithms enhanced the detection system's accuracy and effectiveness.
- Evaluation metrics like Confusion Matrix, ROC Curve, and Precision-Recall Curve provided valuable system performance insights.
- This project significantly contributed to mobile security, mitigating security risks and safeguarding user data and privacy.

Cyberbullying Detection Model | Code Fall2023- LIU

- Developed a Python-based machine learning model using Gradient Boosting, SVM, Random Forest, and Naive Bayes algorithms to detect cyberbullying incidents on social media platforms.
- Achieved high accuracy and efficiency through extensive testing, including accuracy, precision, and F1-score evaluation, as well as confusion matrix analysis. Contributed to creating a safer online environment by proactively identifying and addressing cyberbullying behaviors.

Financial Monitoring System April-2022 - GTU

- I have developed a user-friendly Financial Data Monitoring and Evaluation module using Python, SQLite, HTML, and CSS. Designed to streamline daily expense management for both non-salaried and salaried individuals, the system eliminates manual paperwork and ensures systematic record management.
- Leveraged basic machine learning libraries to enhance functionality and usability. Contributed to creating an accessible and efficient tool for end-users to track and manage their expenses seamlessly.

ACHIEVEMENTS AND CERTIFICATION:

- Runner up in Smart India Hackathon-2020.
- Certification in HTML5 by Michigan University through Coursera.
- Certification in Cybersecurity and the Internet of Things Course by Kennesaw State University through Coursera.
- Certification in Python by Ganpat University.
- Certification in HTML, SQL, and PHP Course in Solo Learn.

PORTFOLIO:

Visit Mehul Parmar's Portfolio for a detailed showcase of projects and certifications.